Tracking discourse status in real time after traumatic brain injury

Kelly Marshall[1,2], Robert Slevc[2], Jared Novick[2], Katherine Sullivan[1], Stefanie Kuchinsky[1]
[1]Walter Reed National Military Medical Center, [2]University of Maryland, College Park
Contact: ksharer@umd.edu

Conversation involves the interplay of social, cognitive, and linguistic processes, dynamically coordinated between partners in real time (e.g., Meyer, 2023). This requires tracking a rapidly changing information structure as conversations unfold, for example, determining whether incoming information is likely to be given (i.e., about things already discussed) or new (e.g., Gardner, 1987). Listeners integrate many cues to interpret given/new discourse status, including speakers' fluency. When an utterance is fluent, listeners expect that a speaker will continue to talk about old information. By contrast, when speakers are disfluent (e.g., "thee uh…"), listeners predict that the speaker will refer to something new (Arnold et al., 2004).

While these findings demonstrate how efficiently healthy listeners use cues to guide incremental interpretation, less is known about whether similar processes operate in individuals with discourse-level difficulties, such as those with traumatic brain injury (TBI; Coelho et al., 2002; Elbourn et al., 2019). Discourse difficulties in TBI have primarily been attributed to deficits in higher-level discourse organization. However, discourse organization measures do not consistently differentiate TBI from healthy groups, suggesting that higher-level measures may not completely account for the cause of patients' difficulty. One possible reason for these mixed results is that traditional discourse measures fail to capture language processing in real time (e.g., Turkstra et al., 2006). TBI is known to disrupt general cognitive operations that support moment-to-moment discourse comprehension, such as prediction, updating expectations, processing speed, and drawing inferences (Levin et al., 2014; MacDonald, 2017). This raises the question of whether individuals with TBI show absent or slowed processes involved in tracking changing discourse information structure, including making and updating predictions about information status. If moment-to-moment comprehension is inefficient, listeners with TBI may miss key information that supports other linguistic, social, or memory processes as discourse continues to unfold. Thus, testing incremental comprehension may help explain additional sources of conversational difficulty following TBI. Understanding sources of breakdown in discourse abilities could also indicate how sentence processing supports typical discourse comprehension.

We employed a discourse task in the visual world paradigm (Arnold et al., 2004). Neurologically healthy adults (N=48) and adults with a history of closed-head TBI (any severity, 6+ months ago; N=23) listened to two-sentence English instructions about where to move four objects. The first sentence established *given* information (e.g., "Move the grapes above the **candle**"). The second sentence mentioned a target that was either given (e.g., candle) or new (e.g., camel). The display contained the target's cohort competitor (e.g., camel for candle). When the target was given, the competitor was new, and vice versa. Half of the trials included a disfluency ("Now move **thee uh** candle…). Generalized additive mixed effects models assessed the effect of information status (given/new, fluent trials), fluency (fluent/disfluent, new trials), and their interaction with group on looks to the competitor, as well as the timing of each effect.

As in Arnold et al. (2004), healthy adults made more predictive looks to the competitor when it was the given item in the fluent condition and the new item in the disfluent condition (Figure 1A). The TBI group showed the same overall pattern (given/new status smooth $b = 5.45$, $p < 0.001$; fluency parametric $b = -0.12$, $p=0.02$; Figure 1B), demonstrating that the use of prior discourse information and of disfluency cues to predict information status is intact in individuals with TBI, at least in simplified discourse contexts. However, the TBI group differed from controls on Disfluent-New trials, where they initially looked less at the (initially unexpected) competitor (interaction smooth $b = 4.24$, $p = 0.02$; Figure 1B). This result may reflect differences in consideration of alternative referents once a prediction has been made. These findings support further examination of temporal dynamics, prediction-updating, and more complex discourse to understand conversation difficulty in TBI (e.g., Kekes-Szabo et al., 2025).

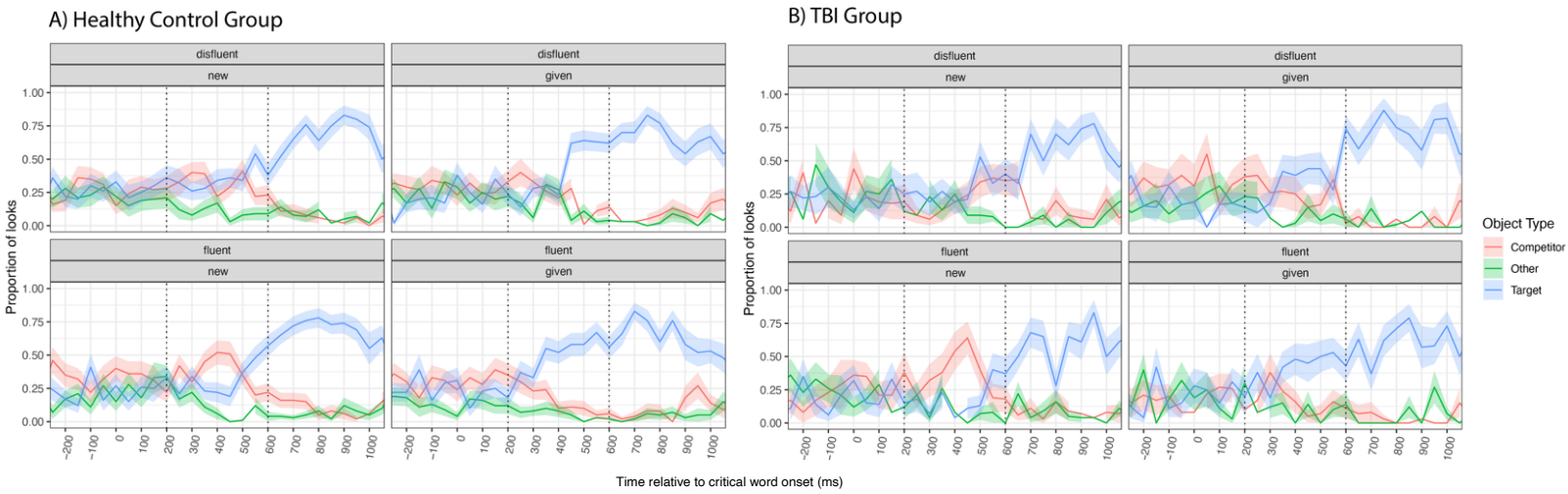

Figure 1. Average proportion of fixations (looks) to each object type per total fixations during each 50-millisecond time slice. Times represent the time in milliseconds since the onset of the critical word (target) within the second instruction. Target = object named, Competitor = unnamed cohort competitor, Other = non-cohort competitor item. Each panel shows responses for trials separated by fluency condition and the information status of the target. Shaded regions represent standard error. Dotted lines show the time window of 200-600 millisecond post target onset, which is the time region of effects described in Arnold et al. (2004). Panel A shows the results for the healthy adult participants (n=48). Panel B shows the results for the TBI patients (n=23).

Disclaimer: The views expressed in this abstract are those of the authors and do not necessarily reflect the official policy of the Department of War or the U.S. Government.

References

Arnold, J. E., Tanenhaus, M. K., Altmann, R. J., & Fagnano, M. (2004). The Old and Thee, uh, New: Disfluency and Reference Resolution. *Psychological Science*, *15*(9), 578–582. https://doi.org/10.1111/j.0956-7976.2004.00723.x

Coelho, C. A., Youse, K. M., & Le, K. N. (2002). Conversational discourse in closed-head-injured and non-brain-injured adults. *Aphasiology*, *16*(4–6), 659–672. https://doi.org/10.1080/02687030244000275

Elbourn, E., Kenny, B., Power, E., & Togher, L. (2019). Psychosocial Outcomes of Severe Traumatic Brain Injury in Relation to Discourse Recovery: A Longitudinal Study up to 1 Year Post-Injury. *American Journal of Speech-Language Pathology*, *28*(4), 1463–1478. https://doi.org/10.1044/2019_AJSLP-18-0204

Gardner, R. (1987). The Identification and role of topic in spoken interaction. *Semiotica*, *65*(1–2). https://doi.org/10.1515/semi.1987.65.1-2.129

Kekes-Szabo, S., Clough, S., Brown-Schmidt, S., & Duff, M. C. (2025). Multiparty Communication: A New Direction in Characterizing the Impact of Traumatic Brain Injury on Social Communication. *American Journal of Speech-Language Pathology*, 1–14. https://doi.org/10.1044/2025_AJSLP-24-00151

Levin, H. S., Shum, D., & Chan, R. C. K. (Eds.). (2014). *Understanding traumatic brain injury: Current research and future directions*. Oxford University Press.

MacDonald, S. (2017). Introducing the model of cognitive-communication competence: A model to guide evidence-based communication interventions after brain injury. *Brain Injury*, *31*(13–14), 1760–1780. https://doi.org/10.1080/02699052.2017.1379613

Meyer, A. S. (2023). Timing in Conversation. *Journal of Cognition*, *6*(1), 20. https://doi.org/10.5334/joc.268

Turkstra, L. S., Brehm, S. E. & Montgomery, E. B. Analysing Conversational Discourse After Traumatic Brain Injury: Isn't It About Time? *Brain Impair.* **7**, 234–245 (2006).